

Data Types

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → "Shanmuka"
- Account Balance → 50000.75
- Account Active → True
- Transaction IDs → [101, 102, 103] → Collection

All these values are different.

Python needs to know:

- Which value is text?
- Which value is a number?
- Which value is a yes/no answer?
- Which value is a collection?

This is where **Data Types** become important.

variables
↑
What kind of data it stores →

Variables:-

name = "Sophie" ←
age = 16 ←
wt = 45.75 ←

Data Types

Single Value

These store one value at a time.

Examples:

- int ✓ → age → 12
- float ✓ → wt → 42.35
- str ✓ → name → "Tejas"
- bool ✓ → is active → false

decimal value

that can change

Collection Data Types

These store multiple values together.

Examples:

- list
- tuple
- set
- dict

you can access by Index

single

collection

Data Type	Example	Mutable/Immutable	Ordered/Unordered	Real-Life Example
int	100	Immutable	N/A → Not Applicable	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name
bool	True	Immutable	N/A	Login Status
list	[1,2,3]	Mutable	Ordered	Shopping Cart
tuple	(1,2,3)	Immutable	Ordered	Coordinates
set	{1,2,3}	Mutable	Unordered	Unique IDs
dict	{'name': 'Rahul'}	Mutable	Insertion-Ordered	Student Record

Comments:-

- It is ignored by Python.
- For telling future self what we are thinking previously.
- For documentation

→ For documentation

Comments

Single line

back ticks
→ ``

Multiline comment

'''

String:-

→ sequence of characters is called string.

OR

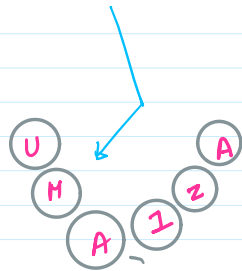
→ Whatever be put inside quote (" ") is called string

Ex:-

```
"Azaan"  
"Umaiza"  
"Nabi"  
"9"  
"123abc# $Azaan"
```

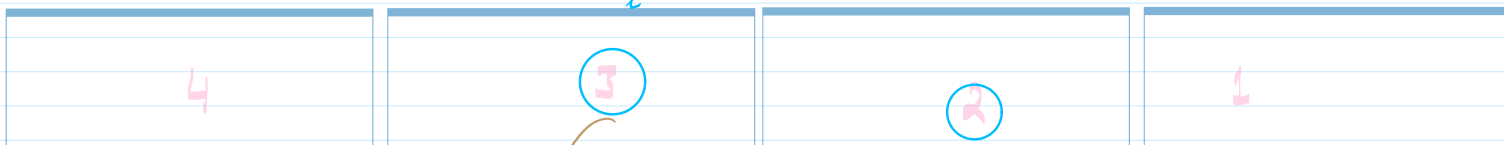
→ Strings are immutable (which cannot be change after creation).

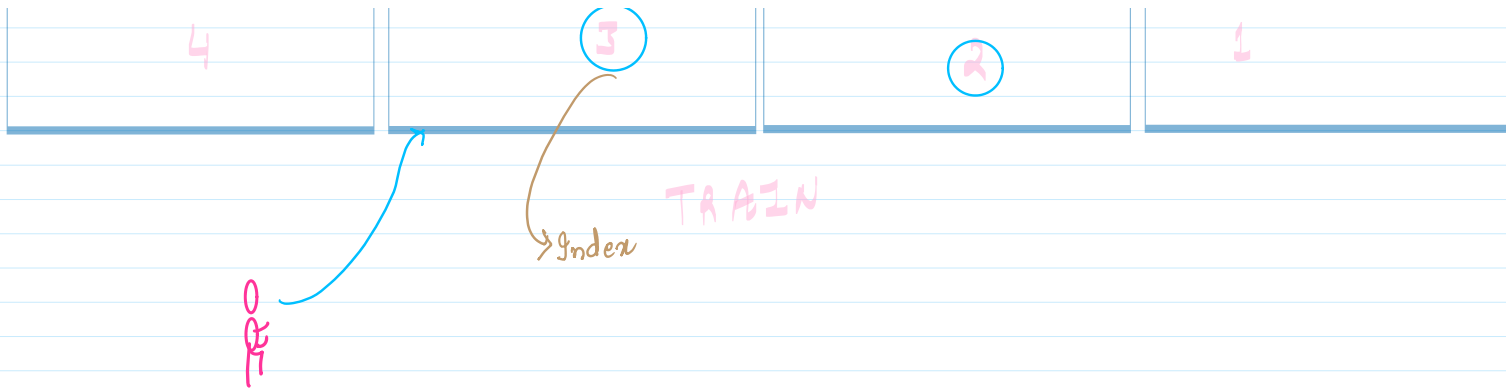
Necklace:-



Indexing:-

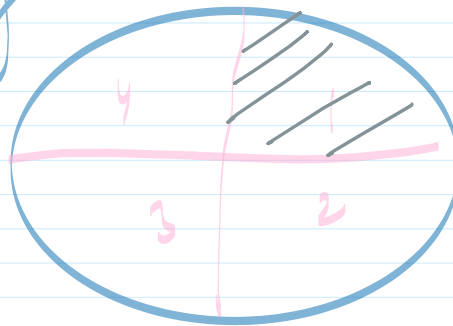
baugies





A Z A A N
 Index → 0 1 2 3 4

String Slicing:-



full-name = A Z A A N S U H A I L
 0 1 2 3 4 5 6 7 8 9 10 11

full-name [startIndex : EndIndex : step]
 ↓ ↓
 included excluded

full-name [3 : 8]